

Do High Credit Limits Causally Affect Default Risk?

0. Abstract

Credit limits play a central role in consumer credit markets, yet economic theory offers opposing predictions: higher limits may induce moral hazard by encouraging riskier borrowing, or they may function as a liquidity buffer, helping households avoid default when facing short-term payment pressure. We assess which mechanism dominates using administrative credit-card data on 30,000 Taiwanese borrowers (April–September 2005). Because banks endogenously assign higher limits to safer customers, we address selection bias using propensity score matching, inverse probability weighting, and doubly robust augmented IPW, conditioning on detailed demographics, repayment histories, and account activity.

We define the treatment discretely as holding a credit limit above the 75th percentile (a “high-limit regime”) compared with all borrowers below this threshold. This specification yields 7,103 well-balanced matched pairs. The AIPW estimate indicates that high-limit borrowers experience a 4.9-percentage-point reduction in next-month default (95% CI: $[-0.0567, -0.0421]$), providing evidence against the moral-hazard hypothesis and in favor of the liquidity-buffer mechanism. Results are robust across alternative thresholds and estimation strategies. Taken together, the evidence indicates that for creditworthy customers, entering a high-limit regime lowers short-run default risk. These findings have direct implications for credit-risk modelling and the design of responsible lending policies.

1. Introduction

Credit limits are a central contractual feature in consumer credit markets. They determine the amount of liquidity available to borrowers, shape consumption smoothing, and serve as an important risk-management tool for lenders. Credit default has long been conceptualized as the outcome of a multi-stage causal process involving borrower fundamentals, macroeconomic shocks, and loan contract terms. The effect of credit limits on default is ambiguous: the moral hazard hypothesis suggests higher limits induce overborrowing (Stiglitz & Weiss, 1981), while the liquidity buffer hypothesis posits they reduce default by providing financial flexibility during income shocks (Gross & Souleles, 2002).

2. Related Work

Empirically, Aydin (2018) finds that limit increases reduce delinquency among subprime borrowers, and Liberman (2016) shows that credit restrictions in Chile raised defaults, both supporting the protective role. However, a central challenge in this literature is causal identification, as banks systematically grant higher limits to less risky borrowers.

Methodologically, Sy (2007) emphasizes that default must be analyzed causally rather than predictively. This study builds on Fahner (2012), who applies propensity score matching to credit decisions, by employing doubly robust AIPW (Bang & Robins, 2005), which remains consistent if either the propensity score or outcome model is correct. We adopt bootstrap variance estimation following Zhang et al. (2025), who show that standard robust errors underestimate uncertainty in matched samples. This causal framework directly informs model risk management: as Kleinberg et al. (2015) argue, while machine learning excels at prediction, policy decisions require causal knowledge.

3. Data and Exploratory Data Analysis (EDA)

The UCI/Kaggle credit default dataset is used, containing demographics, past repayment behavior bill amounts and repayment amounts for 30,000 Taiwanese credit-card holders in 2005. These variables serve as the covariate set for propensity-score estimation. The outcome variable is a binary variable of whether the credit card holder defaults in the next month.

For data cleaning, we remove observations with invalid categories in EDUCATION (values 0, 5, and 6) and MARRIAGE (value 0). Repayment status variables with values below -1 are recoded to -1, reflecting no delay. None of the variables contain missing values. The credit limit distribution of individuals who defaulted and those who didn't differs (Figure A1), and the default rate between treated and controlled group differs by a noticeable amount (Figure A2).

In my model specifically, X includes demographics variables (SEX, AGE, EDUCATION, MARRIAGE), monthly repayment history (PAY_0 to PAY_6), monthly repayment amounts (PAY_AMT1—6), and monthly bill statements (BILL_AMT1—3). BILL_AMT4—6 are excluded as variables are extremely highly correlated (pairwise correlations exceeding 0.90), reflecting the fact that bill balances are carried forward. Including all six variables in a logistic regression propensity score model would introduce severe multicollinearity, leading to unstable coefficient estimates. Thus, only the three most recent months BILL_AMT1–BILL_AMT3 are included for effective confounding control without compromising model stability.

4. Identification Strategy and Methodology

4.1 Identification Strategy

Our goal is to estimate the causal effect of a high credit limit on the probability that a borrower defaults on their credit card in the next month. The challenge is that lenders grant higher limits to borrowers with stronger repayment histories and greater financial stability, the same characteristics that also predict default, leading to selection bias. As a result, we condition on rich pre-treatment covariates and use propensity-score-based estimators (matching, IPW, AIPW) for identification.

Let $Y(1)$ and $Y(0)$ denote the potential outcomes of next month's default under a high or low credit limit respectively. The quantity of interest is the Average Treatment Effect on the Treated (ATT): $ATT = E[Y(1) - Y(0) | T = 1]$. For treatment T , it is a binary variable where $T=1$ if $LIMIT_BAL \geq 75$ th percentile and $T=0$ otherwise. This cutoff is especially chosen based on a balance of sample loss rate after matching and overlap (see later).

Although the underlying credit limit variable is continuous, we adopt a policy-relevant binary treatment estimating the effect of assigning borrowers to a high-limit policy regime rather than estimating a marginal NT\$1 increase in the credit limit. Specifically, we ask the question that among borrowers who currently have high credit limits, how would their probability of default next month change if they had instead been assigned to a lower credit-limit regime. Extending the analysis to estimate the full dose–response function using generalized propensity scores (Imbens, 2000; Hirano & Imbens, 2005) is a natural direction for future work.

4.2 Assumptions for Causal Identification

[1] Ignorability (Conditional Unconfoundedness): $(Y(0), Y(1)) \perp T | X$. This assumption states that, conditional on observed covariates X , the assignment of a high credit limit is statistically independent of potential default outcomes. It makes sense in this context as banks primarily determine credit limits based on observable financial information including past repayment behavior, bill amounts, repayment amounts, and basic demographics. All of these are included in our dataset. While some unobserved factors such as risk preferences may also play a role, the E-value analysis suggests that only a very strong unmeasured confounder could overturn the effect. Thus, the covariates observed here capture the major determinants of credit limit assignment, making the ignorability assumption reasonably credible for this setting.

[2] Overlap: $0 < e(X) < 1$ for all X among treated units. Overlap ensures that treated borrowers have comparable control borrowers with similar covariates. Intuitively, the overlap assumption requires that for any borrower with characteristics X , both high and low credit limits are possible. We examine overlap by plotting the distributions of propensity scores for treated and control borrowers before matching. Across cutoffs between the 55th and 90th percentiles, we find substantial overlap in estimated propensity scores and a high proportion of observations within common support. For the 75th percentile cutoff used in our main specification, the overlap is strong while still yielding a substantively meaningful definition of “high limit”. Observations falling in regions of poor overlap are later discarded by caliper matching.

Under these assumptions, our AIPW estimate of can be interpreted causally as follows: among borrowers who currently receive high credit limits, being assigned to a lower limit would increase their probability of defaulting next month by about the calculated percentage points on average. In

portfolio terms, this corresponds to a substantial change in short-term default risk for the treated segment of the bank's book.

4.3 Propensity Score Estimation

To adjust for differences in borrower characteristics that jointly influence both credit limit assignment and default, we estimate the propensity score defined as the probability of receiving a high credit limit conditional on observed covariates.

We estimate the propensity score using a logistic regression model: $\text{logit}(e(X)) = \beta_0 + \beta_1 \text{SEX} + \beta_2 \text{EDUCATION} + \beta_3 \text{MARRIAGE} + \beta_4 \text{AGE} + \sum_{k \in \{0,2,3,4,5,6\}} \beta_{\text{PAY}_k} \text{PAY}_k + \sum_{j=1}^3 \beta_{\text{BILL}_j} \text{BILL_AMT}_j + \sum_{m=1}^6 \beta_{\text{PAY_AMT}_m} \text{PAY_AMT}_m$. These features collectively serve as proxies for latent credit quality and financial stability, satisfying the ignorability assumption discussed in Section 4.1.

4.4 Matching

To construct a comparable group for borrowers with high credit limits, we implement 1:1 nearest-neighbor propensity score matching (without replacement). This ensures interpretability of the ATT as the average difference in default behavior between otherwise similar individuals receiving different credit limits. The goal of matching is to pair each treated unit (high-limit borrower) with a control unit (low-limit borrower) who is similar in observable characteristics X , thereby approximating a randomized experiment. Each treated unit is matched to the nearest control unit with the smallest absolute difference in propensity scores.

To prevent inappropriate matches between borrowers with significantly different propensity scores in regions of poor overlap, we impose a caliper of 0.2 standard deviations of the logit propensity score, following the recommendation of Rosenbaum and Rubin (1985). This restriction ensures that a treated borrower is matched only to control borrowers whose estimated likelihood of receiving a high credit limit by the logit of the propensity score is sufficiently similar (within 0.2 SD), and observations outside this are discarded.

The matching estimator is $\widehat{ATT}_{\text{match}} = \frac{1}{N_T} \sum_{i:T_i=1} (Y_i - Y_{j(i)})$ where N_T is the number of treated units retained after matching, and $j(i)$ denotes the matched control unit for treated unit i . This estimator is unbiased under the ignorability and overlap assumptions. The propensity score overlap after matching improves significantly (Figures A3, A4).

4.5 Inverse Probability Weighting (IPW)

IPW reweights the observed data to create a sample in which the distribution of covariates is independent of treatment assignment. Instead of selecting a subset of borrowers as in matching, IPW uses all observations but reweights (increases the weighting of) the controlled group who resemble the treated group based on their estimated propensity scores.

Let $e(X_i) = \Pr(T_i = 1 | X_i)$ denote the estimated propensity score. For ATT estimation, the IPW weights are: $w_i = \begin{cases} 1, & T_i = 1 \\ \frac{e(X_i)}{1-e(X_i)}, & T_i = 0 \end{cases}$, $\widehat{ATT}_{\text{IPW}} = \frac{1}{N_T} \sum_{i:T_i=1} Y_i - \frac{1}{N_T} \sum_{i:T_i=0} w_i Y_i$.

Thus, control units with higher propensity score receive higher weights.

4.6 Augmented Inverse Probability Weighting (AIPW)

The AIPW estimator combines propensity score reweighting with the outcome model. It is doubly robust as the estimator is consistent if either the propensity score model or the outcome regression model is correctly specified.

In addition to the propensity score model, we estimate the following outcome regression model:
 $\hat{\mu}_t(X_i) = \hat{E}[Y | T = t, X_i], t \in 0,1$. We fit a logistic regression of default on treatment and covariates, and obtain $\mu^t(X_i) = Pr(Y = 1 | T = t, X_i)$ by setting $T=t$ in the same fitted model.

The ATT is estimated by $ATT_{AIPW} = \frac{1}{N_T} \sum_{i:T_i=1} [Y_i - \hat{\mu}_0(X_i)] - \frac{1}{N_T} \sum_{i:T_i=0} \frac{\hat{e}(X_i)}{1-\hat{e}(X_i)} [\hat{\mu}_1(X_i) - Y_i]$. This model provides an estimate of how likely a borrower is to default, conditional on covariates and treatment status. Thus, this answers our question of how the borrowers' default risk change when it is given a higher credit limit (treated) compared to a lower one.

4.7 Mechanism Analysis via Causal Mediation

To better understand how higher credit limits reduce default risk, we conduct a causal mediation analysis (Imai et al., 2010). For each mediator M_{ik} , we estimate the Average Causal Mediation Effect (ACME), the Average Direct Effect (ADE), and the proportion of the total effect explained by the mediator. Mediator model is $M_{ik} = \alpha_0 + \alpha_1 \text{treat}_i + \alpha^\top X_i + \varepsilon_i$; and outcome model is $Y_i = \beta_0 + \beta_1 \text{treat}_i + \beta_2 M_{ik} + \beta^\top X_i + u_i$

Four mediators are considered, each motivated by economic mechanisms related to liquidity constraints and repayment pressure: M1 Balance Ratio (BILL_AMT1 / LIMIT_BAL) capturing utilization and short-term liquidity tightness.; M2 Bill Growth (BILL_AMT1 – BILL_AMT2) reflecting month-to-month spending shocks; M3 Repayment Burden (PAY_AMT1 / LIMIT_BAL) representing the relative size of scheduled payments to available credit; and M4 Available Credit (LIMIT_BAL – BILL_AMT1) a direct measure of liquidity slack or buffer capacity.

We estimate ACME and ADE for each mediator using nonparametric bootstrap (1,000 replications), adjusting for demographic and credit-history covariates to satisfy the sequential ignorability assumption required for causal identification.

5. Results

5.1 Covariates Balance After Matching

As Figure 1 below depicts, after nearest-neighbor matching with a caliper of 0.2, covariate balance improves substantially. All SMDs fall near zero and well below 0.1. This confirms that the matched treated and control groups are highly comparable on observed characteristics, providing a credible basis for causal effects analysis.

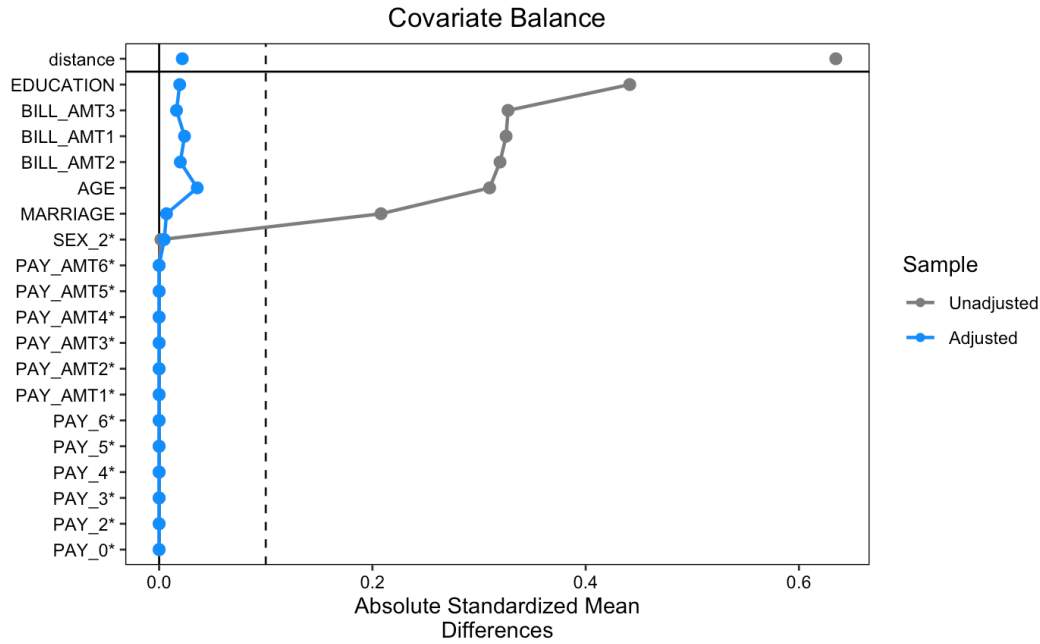


Figure 1. Standardized mean differences (SMDs) for all covariates before (grey) and after (blue) propensity score matching. The vertical dashed line denotes the conventional 0.1 threshold for acceptable balance

5.2ATT

To examine the robustness of the causal effect to different treatment definitions, we estimate the ATT across a range of credit-limit cutoffs (55%–90% quantiles of the limit distribution, as to make the treated group larger than the controlled group but not too extreme). Table 1 summarizes the results for the different cutoffs.

Cutoff	Treated Share	Matched Pairs	Loss Rate	ATT(Match)	ATT(IPW)	ATT(AIPW)
55%	0.453	10752	0.274	-0.121	-0.102	-0.003
60%	0.412	10265	0.306	-0.114	-0.098	-0.015
65%	0.372	9728	0.343	-0.117	-0.098	-0.028
70%	0.321	8732	0.410	-0.111	-0.097	-0.038
75%	0.257	7103	0.520	-0.106	-0.095	-0.049
80%	0.207	5834	0.606	-0.106	-0.095	-0.064
85%	0.153	4324	0.708	-0.102	-0.096	-0.071
90%	0.112	3238	0.781	-0.096	-0.093	-0.073

Table 1. Estimated ATT effects across alternative credit-limit thresholds (55%–90%), showing treated shares, matched sample sizes, loss rates, and ATT estimates from matching, IPW, and AIPW.

Across all cutoffs, the matching estimator yields a consistently negative ATT, indicating that borrowers with higher credit limits exhibit lower default rates compared to similar borrowers with lower limits. The IPW estimator produces nearly identical patterns, with estimates slightly closer to zero, likely because weighting smooths residual imbalance rather than discarding unmatched units.

In contrast, the AIPW estimator produces negative effects of smaller magnitude, especially at lower cutoffs. This is expected as default is a sparse binary outcome and covariates such as payment delinquency and bill amounts are highly predictive, the outcome regression component of AIPW tends to shrink residuals toward zero. When combined with the propensity-score weighting term, this leads to more conservative estimates.

We use the 75% cutoff as our primary specification, as it offers a balance between adequate treated sample size and acceptable loss rate of unmatched pairs. At this cutoff, the AIPW estimator yields an ATT of -0.049 , implying that moving from a low to a high credit limit reduces default probability by approximately 4.9%.

To yield the confidence interval of the ATT estimates, bootstrap inference using 5,000 replications shows that the matching estimator ATT has a 95% CI of $[-0.1190, -0.0937]$ under the 75th-percentile cutoff, the IPW estimator CI is $[-0.1089, -0.0831]$, and AIPW gives CI $[-0.0567, -0.0421]$. All confidence intervals lie strictly below zero, confirming the conclusion that higher credit limits causally reduce default risk.

As a robustness check, interactions between treatment and (i) repayment delinquency indicators (PAY_0 to PAY_6), (ii) recent bill amounts (BILL_AMT1–3), (iii) recent repayment amounts (PAY_AMT1–3), and (iv) age are included, which allow the causal effect of credit limits to vary flexibly across borrower characteristics related to repayment behavior and liquidity pressure. The resulting ATT remains virtually identical to the baseline estimate (-0.0505 with a 95% bootstrap CI of $[-0.0609, -0.0394]$, compared to -0.0490 with $[-0.0565, -0.0418]$ in the no-interaction model). The negligible change in magnitude and overlapping confidence intervals indicate minimal treatment effect heterogeneity and confirm that the causal effect of credit limits is not sensitive to outcome-model specification. This suggests that higher credit limits reduce default risk and that this effect operates primarily through a stable liquidity buffer mechanism rather than subgroup-specific responses (will be confirmed by mediation analysis later as well).

5.3 Sensitivity Analysis

5.3.1 Rosenbaum Sensitivity Test

To evaluate the method of matching, Rosenbaum sensitivity test shows that the upper-bound p-value remains below 0.05 for gamma up to 1.8. This means an unobserved confounder would need to increase the odds of receiving a high credit limit by roughly 80% between two otherwise identical borrowers in order to explain away the estimated effect. Lower-bound p-values equal to zero are expected in sign tests with sparse discordant pairs and do not indicate hidden bias. Only at gamma = 1.9 does the result lose significance at 5% level, indicating moderate robustness to unobserved confounding.

However, the Rosenbaum test has limited statistical power in this setting because the outcome (default) is binary and relatively rare. Among the 7,103 matched pairs, only 31.8% are discordant pairs that contribute information to the sign test. For this reason, I additionally report E-values, which are based on risk-ratios rather than within-pair outcome differences, and thus retain power even when discordant pairs are sparse.

5.3.2 E-Value

To further testing the model's sensitivity to unobserved confounding, as the matching and weighting methods remove bias for observed covariates, the Cornfield-style sensitivity analysis is employed using the E-value framework proposed by VanderWeele and Ding (2017). This E-value assesses how strong an unmeasured confounder would need to be to explain away an observed association.

The estimated risk ratio (RR) of default for treated vs. control borrowers is $RR = \frac{P(Y=1|T=1)}{P(Y=1|T=0)}$. The Cornfield inequality states that for an unmeasured confounder U to fully explain away the observed association, RR_{UY} (confounder-outcome association) and RR_{UT} (confounder-treatment association) abide towards the following inequality: $RR_{UY} \times RR_{UT} \geq \frac{1+\sqrt{1-RR}}{RR}$.

The E-value simplifies to $E = RR^{-1} + \sqrt{RR^{-1}(RR^{-1} - 1)}$, which represents the minimum strength of association that U must have with both treatment and outcome to explain away the treatment effect.

The result of our model gives an $RR = 0.562$, and $E = 2.96$, which means that an unmeasured confounder would need to increase the risk of receiving a high credit limit and the risk of default by at least 2.96-fold each to explain away the observed effect.

Bootstrapping this E-value on the matched dataset, we get an upper bound of 2.68 at 95% confidence interval, which suggests that the model is robust and does not suffer from unobserved confounding factors as few plausible unmeasured factors are likely to have associations that large with both treatment assignment and default.

5.4 Robustness Checks

5.4.1 Negative Outcome (Placebo) Tests

To further assess whether the estimated treatment effect could arise from model misspecification or unobserved confounding, placebo tests are conducted using SEX and AGE as pseudo-outcomes, which are variables that credit limits cannot plausibly affect. Re-estimating the ATT with the same matching procedure and conducting t-tests yields effects that are statistically indistinguishable from zero (SEX: $p=0.093$, 95% CI: $[-0.0022, 0.0288]$; AGE: $p=0.668$, 95% CI: $[-0.22, 0.35]$). These null placebo results confirm that the procedure behaves appropriately when no effect is expected, reinforcing confidence in the validity of the main causal estimate.

5.4.2 Heterogenous Treatment Effects

To assess whether the effect of higher credit limits varies more for certain sub-groups of borrowers, I examine treatment heterogeneity by baseline credit risk and gender. Baseline risk is proxied by the estimated propensity score, which reflects the bank's pre-treatment assessment of borrower creditworthiness. Borrowers are stratified into three groups ($pscore < 0.10$, $0.10-0.20$, ≥ 0.20), following empirical guidance on propensity-score stratification for credit risk data (Karlan & Zinman, 2009; Agarwal et al., 2018) and the distribution of my sample, where the 25th, 50th, and 75th percentiles of the propensity score are 0.148, 0.227, and 0.323 respectively. ATT for each subgroup is computed using the IPW estimator to ensure covariate balance within each stratum.

The results show clear differences in magnitude but a consistent pattern in direction. Low-risk borrowers exhibit a modest effect (ATT = -0.025 ; 95% CI: -0.078 to 0.024 ; sample size $N = 2,351$), while the treatment is substantially larger for medium-risk borrowers (ATT = -0.149 , 95% CI: -0.175 to -0.123 ; $N = 9,519$), suggesting that the liquidity-buffer effect is strongest for this segment. High-risk borrowers also benefit (ATT = -0.086 ; 95% CI: -0.102 to -0.071 ; $N = 17,731$) but to a lesser extent, consistent with chronic credit problems limiting the impact of additional liquidity. For subgroup analysis based on gender, differences are mild: men (ATT = -0.112 ; 95% CI: -0.133 to -0.090 ; $N = 11,746$) and women (ATT = -0.085 ; 95% CI: -0.101 to -0.069 ; $N = 17,855$) both show beneficial effects, with no sign reversals. This indicates that higher credit limits consistently reduce default risk across all examined subgroups, with the strongest effect concentrated among medium-risk, liquidity-constrained borrowers.

5.5 Mediation Analysis

Mediator	ACME	ADE	Total Effect	Prop Mediated
M1 Balance Ratio	-0.0131	-0.0967	-0.1057	12.37%
M2 Bill Growth	-0.0008	-0.1057	-0.1062	0.78%
M3 Repayment Burden	-0.0348	-0.0775	-0.1037	33.53%
M4 Available Credit	-0.0873	-0.0319	-0.1103	79.13%

Table 2. Causal Mediation Decomposition of the Effect of Credit Limits on Default. Average Causal Mediation Effect (ACME), the Average Direct Effect (ADE), the total effect, and the share of the treatment effect explained by each mediator are estimated based on 1,000 bootstrap replications. ACME quantifies the portion of the treatment effect transmitted through the mediator, while ADE captures the remaining direct pathway.

Across all specifications, shown in Table 2, the mediation analysis shows that higher credit limits reduce default primarily by easing short-term liquidity pressure, rather than by affecting long-term repayment behavior or exposure to income shocks.

M1 (Balance Ratio) explains about 12% of the total effect, reflecting the mechanical reduction in utilization when credit limits increase. M2 (Bill Growth) contributes almost no mediation (<1%), suggesting that month-to-month bill shocks are not a major pathway through which credit limits affect default. M3 (Repayment Burden) mediates about 33% of the effect, indicating that reduced repayment pressure is a meaningful channel. M4 (Available Credit) captures the liquidity slack channel and mediates the largest share (79%) of the effect. This suggests that borrowers with unused credit can withstand short-term liquidity shocks and postpone urgent repayment, substantially reducing the probability of default.

Taken together, these results suggest that liquidity availability instead of income shocks or bill growth is the central mechanism linking high credit limits to lower immediate default risk. Credit limits shape borrowers' immediate liquidity positions and the tightness of their repayment constraints. Increasing a borrower's limit expands available liquidity, alters their intertemporal budget set, and reduces exposure to short-term liquidity risk, lowering the likelihood of default.

6. Discussion & Future Recommendations

In conclusion, across all identification strategies, we find that borrowers with credit limits in the top quartile face substantially lower next-month default risk compared with observationally similar borrowers. This short-run protective effect aligns with theories in which higher limits alleviate short-term liquidity constraints, allowing borrowers to smooth payments and avoid delinquency. Our ATT estimate captures this effect on the risk-difference scale, while the E-value based on the matched risk ratio indicates that a confounder would need to be associated with both treatment assignment and default risk by nearly threefold to explain away the effect.

However, several limitations qualify our conclusions. First, the outcome is one-month default, so our estimates capture only immediate credit-risk responses to high limits, which longer-run effects remain unexamined. Second, although we condition on rich pre-treatment financial variables, credit limits are complex and partially proprietary lender decisions, leaving open the possibility of residual unmeasured confounding. Finally, our binary treatment definition reflects a policy-relevant threshold, but credit limits are inherently continuous, and the response to limits may be nonlinear.

Future work could therefore pursue three directions. First, incorporating richer borrower-level information, such as income, credit scores, and external debt obligations would strengthen ignorability and further contextualize the E-value's robustness claims. Second, extending the analysis to continuous-treatment methods, such as generalized propensity scores, would allow estimation of a full dose-response curve for credit limits and inform optimal limit-setting policies. Lastly, modeling multi-period outcomes using survival analysis or marginal structural models would provide insight into the medium-long-run consequences of credit-limit policies, bridging the gap between immediate liquidity effects and longer-term financial behavior. Together, these extensions would deepen our understanding of how credit limits shape borrower outcomes across time and across the credit-risk spectrum.

Bibliography

- Agarwal, S., Chomsisengphet, S., Mahoney, N., & Stroebe, J. (2018). Do banks pass through credit expansions to consumers who want to borrow? *Quarterly Journal of Economics*, 133(1), 129–190.
- Austin, P. (2011). Optimal caliper widths for propensity-score matching. *Pharmaceutical Statistics*, 10(2), 150–161.
- Aydin, D. (2018). The marginal propensity to consume over the business cycle. *American Economic Journal: Macroeconomics*, 10(2), 351–384.
- Bang, H., & Robins, J. M. (2005). Doubly robust estimation in missing data and causal inference models. *Biometrics*, 61(4), 962–973.
- Fahner, Y. (2012). Estimating Causal Effects of Credit Decisions. Tilburg University.
- Fong, C., Hazlett, C., & Imai, K. (2018). Covariate balancing generalized propensity score. *Journal of the Royal Statistical Society: Series B*, 80(4), 719–744.
- Gross, D. B., & Souleles, N. S. (2002). Do liquidity constraints and interest rates matter for consumer behavior? Evidence from credit card data. *Quarterly Journal of Economics*, 117(1), 149–185.
- Hirano, K., & Imbens, G. W. (2005). The Propensity Score with Continuous Treatments. In *Applied Bayesian Modeling and Causal Inference from Incomplete-Data Perspectives: An Essential Journey with Donald Rubin's Statistical Family* (pp. 73–84). Wiley-Blackwell.
- Imai, K., Keele, L., & Tingley, D. (2010). A general approach to causal mediation analysis. *Psychological Methods*, 15(4), 309–334.
- Imbens, G. W. (2000). The Role of the Propensity Score in Estimating Dose-Response Functions. *Biometrika*, 87(3), 706–710. <http://www.jstor.org/stable/2673642>
- Karlan, D., & Zinman, J. (2009). Observing unobservables: Identifying information asymmetries with a consumer credit field experiment. *Econometrica*, 77(6), 1993–2008.
- Kleinberg, J., Ludwig, J., Mullainathan, S., & Obermeyer, Z. (2015). Prediction policy problems. *American Economic Review*, 105(5), 491–495.
- Liberman, A. (2016). The value of a good credit reputation: Evidence from credit card renegotiations. *Journal of Financial Economics*, 120(3), 644–660.
- Matcham, W. (2025). Risk-Based Borrowing Limits in Credit Card Markets. SSRN Working Paper. <https://doi.org/10.2139/ssrn.4926974>
- Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1), 41–55.
- Rosenbaum, P. R., & Rubin, D. B. (1985). Constructing a control group using multivariate matched sampling methods that incorporate the propensity score. *The American Statistician*, 39(1), 33–38.
- Rubin, D. B. (1974). Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology*, 66(5), 688–701.
- Stiglitz, J. E., & Weiss, A. (1981). Credit rationing in markets with imperfect information. *American Economic Review*, 71(3), 393–410.

Sy, W. (2007). A Causal Framework for Credit Default Theory. Australian Prudential Regulation Authority.

VanderWeele, T. J., & Ding, P. (2017). Sensitivity analysis in observational research: Introducing the E-value. *Annals of Internal Medicine*, 167(4), 268–274.

Zhang, Z., Kim, H. J., Lonjon, G., & Zhu, Y. (2025). Balance diagnostics after propensity score matching. *Annals of Translational Medicine*, 7(1), 16.

Appendix

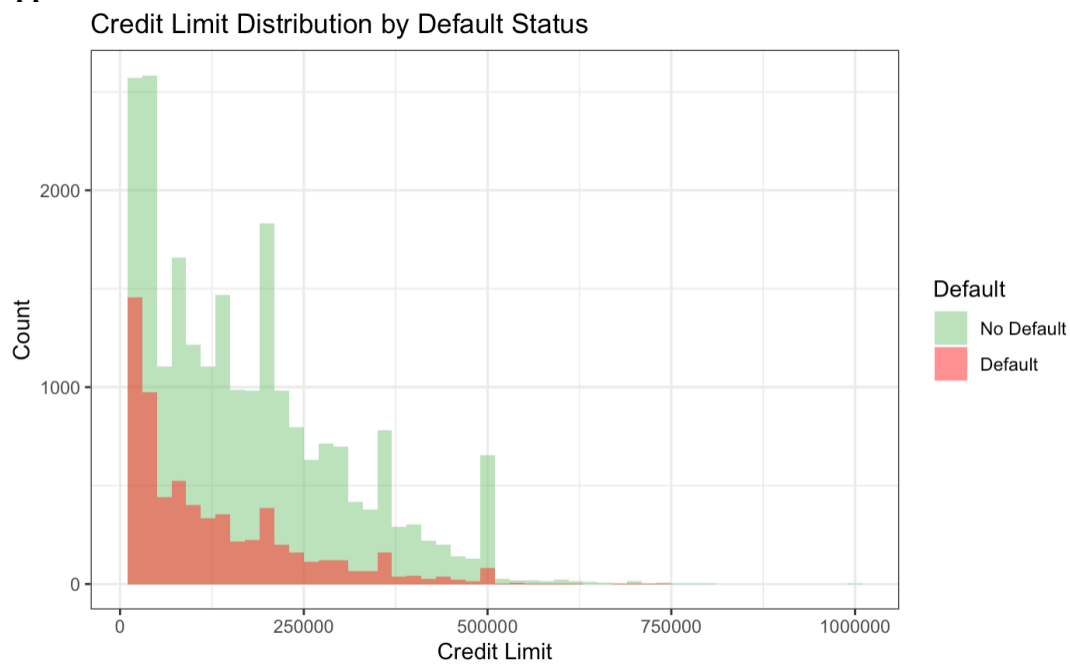


Figure A1. Distribution of credit limits by default status. Credit limits range from NT\$10,000 to NT\$1,000,000 with a highly right-skewed shape. Borrowers who eventually default tend to have lower credit limits, reflecting banks' ex-ante risk assessments.

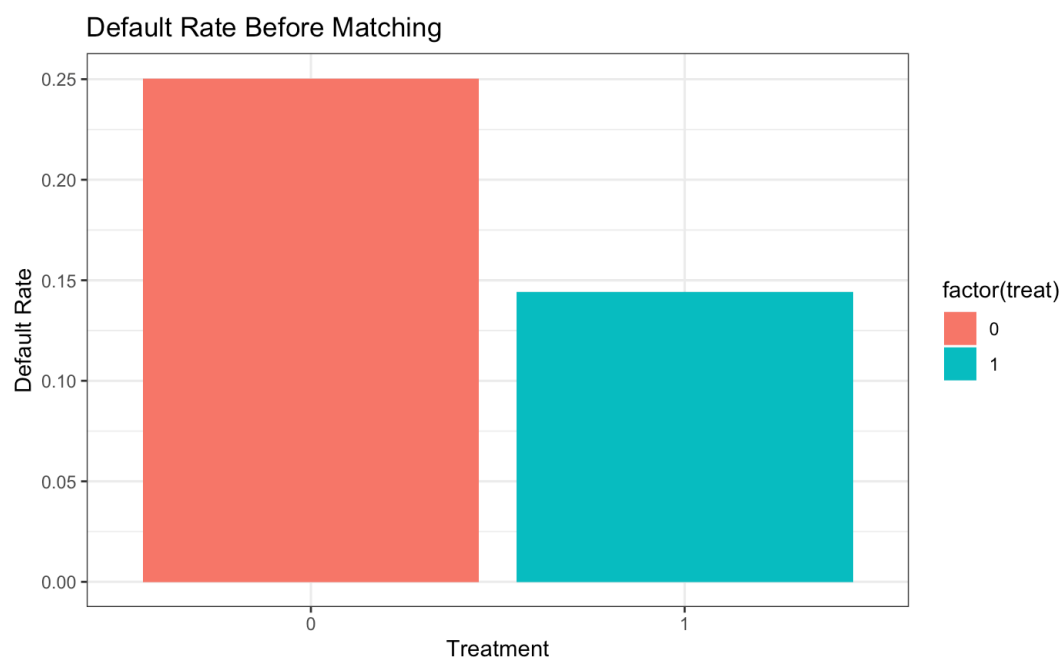


Figure A2. Default rates by treatment status before matching. The treated group (high credit limits) has a substantially lower default rate (14.2%) than the control group (22.7%), indicating strong selection bias and motivating propensity-score adjustment.



Figure A3. The propensity score overlap for the 75th-percentile treatment definition (main specification) before matching

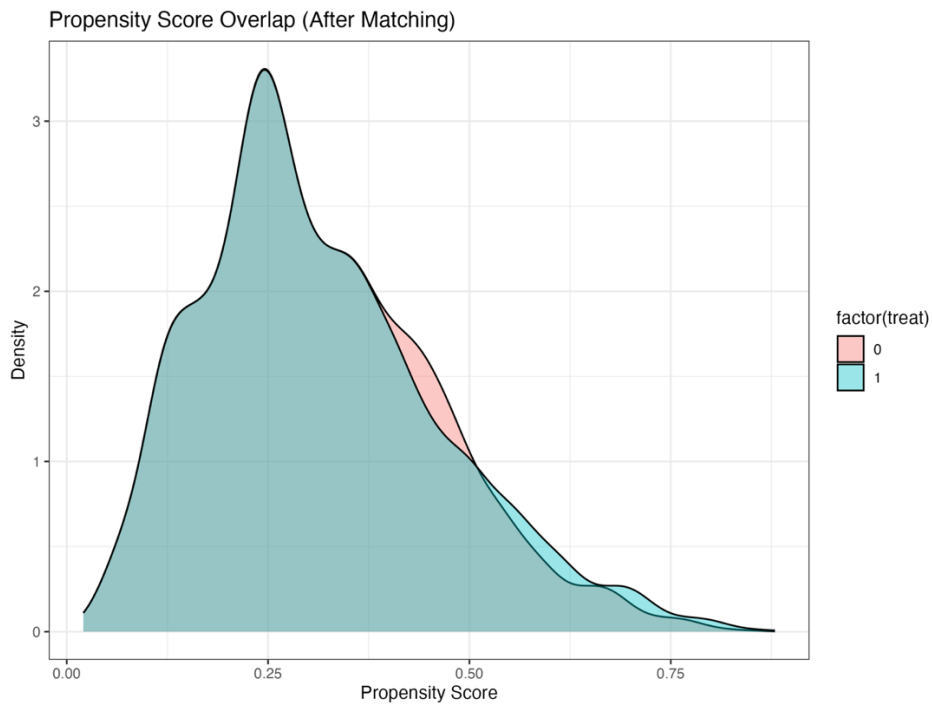


Figure A4. The propensity score overlap for the 75th-percentile treatment definition (main specification) after matching